



Let me analyze this hypothesis-experiment pair step by step.

Hypothesis Assessment: The hypothesis has two clear claims: (1) Swin Transformer captures better global visual-language features than CNNs for ZS3, and (2) pixel-text score maps with modified decision boundaries reduce seen-bias and improve unseen class performance. The hypothesis is specific and testable, though not particularly novel \u2014 using transformers over CNNs is a relatively straightforward architectural upgrade, and the seen-bias problem is well-known in zero-shot learning.

Experiment Assessment: Experiment 1 & 2: These evaluate the full framework on two standard benchmarks (PASCAL VOC and PASCAL Context) with multiple seen/unseen splits. They compare against relevant baselines (DeViSE, SPNet, ZS3Net, CSRL, JoEm). The metrics (mIoUs, mIoUu, hIoU) are appropriate and standard for ZS3 evaluation. Multiple splits provide robustness to the evaluation. This is solid experimental design. Experiment 3: The ablation study is well-designed \u2014 it isolates the two main contributions (transformer backbone vs. CNN, and the auxiliary score map loss) by testing combinations. This directly addresses both parts of the hypothesis.

Strengths:- Clear, testable hypothesis- Multiple benchmarks and splits- Appropriate baselines from the literature- Standard and appropriate metrics including harmonic mean (which captures the seen-unseen trade-off)- Ablation study that isolates individual contributions

Weaknesses:- Only uses swin-tiny; no exploration of other transformer sizes or variants- The ablation is only on one split (unseen-6 of PASCAL Context), which limits generalizability of ablation conclusions- No statistical significance tests or variance reporting mentioned- Missing some more recent baselines potentially- Only two datasets, both from PASCAL family \u2014 no COCO-stuff or other diverse benchmarks- The novelty is somewhat incremental (replacing CNN with Swin Transformer + an auxiliary loss)

Overall, the **experimental design is reasonably rigorous with proper baselines, ablations, and standard metrics across multiple settings**. **The methodology is sound**, though could be more comprehensive. **The ablation study directly addresses the hypothesis components**. I'd classify this as leaning **toward high rigor**. The experiments are controlled, include appropriate baselines and ablations, and use suitable metrics, even if there are minor gaps.